

Dr. Jianghao Liu

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[Google Scholar](#) (Citation within 5 years: **750**; indice h: 13)



SELECTED PUBLICATION (< 5 years)

First author:

1. **Liu, J.** (2026) A neural model of conscious mental imagery and aphantasia. *Neuropsychologia*.
2. **Liu, J.**, & Bartolomeo, P. (2025). Aphantasia as a functional disconnection. *Trends in Cognitive Sciences*.
3. **Liu, J.**, Zhan, M., Hajhajate, D., Spagna, A., Dehaene, S., Cohen, L., & Bartolomeo, P. (2025). Visual mental imagery in typical imagers and in aphantasia: A millimeter-scale 7-T fMRI study. *Cortex*
4. **Liu, J.**, Bayle, D. J., Spagna, A., Sitt, J. D., Bourgeois, A., Lehongre, K., Fernandez-Vidal, S., Adam, C., Lambrecq, V., Navarro, V., Seidel Malkinson, T., & Bartolomeo, P. (2023). Fronto-parietal networks shape human conscious report through attention gain and reorienting. *Communications Biology*, 6(730).
5. **Liu, J.**, Cohen, L., Palier, C., Dehaene, S., Zhan, M., & Bartolomeo, P. (2025). Face-, color-, and word-specific patches in the human orbitofrontal cortex. bioRxiv, 2025-04. (Under review)
6. **Liu, J.**, & Bartolomeo, P. (2023). Probing the unimaginable: The impact of aphantasia on distinct domains of visual mental imagery and visual perception. *Cortex*, 166, 338–347.
7. **Liu, J.**, Spagna, A., & Bartolomeo, P. (2022). Hemispheric asymmetries in visual mental imagery. *Brain Structure and Function*, 227(2), 697–708.

Senior author:

8. Scholz, C. O., Monzel, M., & **Liu, J***. (2025). Absence of shared representation in the visual cortex challenges unconscious imagery in aphantasia. *Current Biology*.
9. Scholz, C. O.*, Monzel, M., Kvamme, T., **Liu, J***, & Silvanto, J. (2026). An Integration Model of Mental Imagery and Aphantasia: Conceptual Framework, Neuromechanistic Pathways, and Clinical Implications *Neuropsychologia*

Coauthors:

10. Bartolomeo, P., **Liu, J.**, & Malkinson, T. S. (2024). Frontoparietal asymmetries leading to conscious perception. *Trends in Cognitive Sciences*.
11. Seidel Malkinson, T., Bayle, D. J., Kaufmann, B. C., **Liu, J.**, Bourgeois, A., Lehongre, K., ... & Bartolomeo, P. (2024). Intracortical recordings reveal vision-to-action cortical gradients driving human exogenous attention. *Nature Communications*, 15(1), 2586.
12. Bartolomeo, P., Hajhajate, D., **Liu, J.** & Spagna, A. (2020) Assessing the causal role of early visual areas in visual mental imagery. *Nature Reviews Neuroscience*
13. Spagna, A., Hajhajate, D., **Liu, J.** & Bartolomeo, P. Visual mental imagery engages the left fusiform gyrus, but not the early visual cortex: A meta-analysis of neuroimaging evidence. *Neuro & Biobeh Reviews* (2021).

COMMUNITY BUILDER

1. Founder of a research club on Aphantasia: **IRCA** (Interdisciplinary Reading Club of Aphantasia), [link](#)
2. Co-organiser of the international **IRCA conference** on aphantasia (2026/09, Glasgow), [link](#)
3. Chair of the symposium “Aphantasia and Consciousness” in **ASSC** conference (2026/07, Chile), [link](#)
4. New report by **Le Monde** (2024): “Vivre sans avoir d’images mentales... et ne pas le savoir”, [link](#)
5. Interview in **Nature** (2026): “Many people have no mental imagery. What’s going on in their brains?”, [link](#)

AWARD

1. “European Workshop on Cognitive Neuropsychology Prize”, Early career award, 2023, Bressanone, Italy.
2. “SAW 2025 - Seeing and Acting Workshop”, Best abstract award, 2025, Coimbra, Portugal.
3. “Next frontiers in consciousness research”, Travel grant, 2023, NIH, Bethesda, United States.
4. “Carnot training grant” for visiting scholar in Columbia University in NYC, 2023, ICM, France.

SCIENTIFIC COMMUNICATIONS

1. European Workshop on Cognitive Neuropsychology, Bressanone, Italy; 2022, 2023, 2024
2. Next frontiers in consciousness research workshop, 2023, NIH, Bethesda, United states.
3. ASSC consciousness, 2023 (New York), 2024 (Tokyo).
4. HBP Symposium: From Cortical Microcircuits to Consciousness, 2022, Paris, France.
5. FENS Forum of Neuroscience, 2022, Paris, France.
6. Forum des Science Cognitive, 2025, Nancy

EXPERIENCE

Postdoctoral researcher in cognitive neuroscience

Paris Brain Institute (ICM) & NeuroSpin Center

Dec.2023 - Now

Paris, France

Visiting scholar

Columbia University, Zuckerman brain institute

Sep.2023 –Nov.2023

New York, USA

Corporate Science & Research, Dassault Systèmes

Permanent contract, Living brain technology specialist

2018 – now

Paris, France

EDUCATION

PhD Candidate in cognitive neuroscience

PICNIC lab, Institut du Cerveau – Paris Brain Institute (ICM)

Sep.2019 -Dec.2023

Paris, FR

MSc. Bioengineering in Neuroscience

ParisTech – ESPCI & Université Paris Descartes

Sep.2017- Oct.2018

Paris, FR

Engineering diploma in Mechanics

Arts et Métiers ParisTech (ENSAM)

Sep.2015- Oct.2018

Paris, FR

LANGUAGES

French Bilingual

Anglish Bilingual

Chinese Native